

Homework — 1.1.3 Input, Output and Storage Devices

Name: _____ Date: _ **Mark:** ____ / 20

OCR H446 — set after the 1.1.3 lesson. Show your working.

Part A — This week's topic (~14 marks)

1. For each device, state whether it is an **input** or **output** device. [3] (AO1)

- Microphone: _____
- Laser printer: _____
- Sensor: _____

2. Describe how data is stored and read on an **optical** disc such as a DVD. [2] (AO1)

3. A laptop manufacturer replaces a magnetic **HDD** with a **flash (SSD)** drive. State **two** advantages and **one** disadvantage of this change. [3] (AO1)

Advantages: _____

Disadvantage: _____

4. A museum needs to **archive** several terabytes of footage that will rarely be accessed again, at the **lowest possible cost**. Recommend a suitable storage medium and **justify** your choice against the requirements. [3] (AO2)

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5. Explain the difference between **RAM** and **cache** in terms of speed and location. [2] (AO1)

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6. A user complains their computer becomes extremely slow when many large programs are open at once, and the hard disk activity light stays on constantly. Using the term **virtual memory**, explain what is happening. [1] (AO2)

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Part B — Retrieval: keep it fresh (~6 marks)

7. State **two** differences between **RISC** and **CISC** processors. [2] (retrieval from 1.1.2)

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8. Explain why a **GPU** is well suited to applying the same calculation to a large data set. [2] (retrieval from 1.1.2)

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9. State the **full name** and **role** of the **MDR**. [2] (retrieval from 1.1.1)

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